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DOPPLER PROCESSING IN FREQUENCY-MODULATED CONTINUOUS WAVE RADAR SYSTEMS USING DITHER

Abstract

Systems, methods and non-transitory computer readable mediums are provided for processing radar data to improve detection of weaker targets in the presence of stronger targets. One such system comprises a radar sensor to receive radar signals; and a processor to generate a Doppler Fourier Transform (Doppler-FT) representation of the radar signals, and perform processing to adjust a noise floor in the Doppler-FT representation from a first level to a second level that is lower than the first level. The processing to adjust the noise floor is based on a velocity associated with a first signal peak visible in the Doppler-FT representation when the noise floor is at the first level, and a second signal peak not visible in the Doppler-FT representation when the noise floor is at the first level is visible in the Doppler-FT representation when the noise floor is at the second level.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of, and claims priority to, application Ser. No. 18/401,920, filed Jan. 2, 2024, which is a continuation of, and claims priority to, application Ser. No. 17/132,462, filed Dec. 23, 2020 (now U.S. Pat. No. 11,899,095), which claims priority to India application 202041040230, filed Sep. 17, 2020, the content of each of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This invention relates to electrical systems, and more particularly, to Doppler processing for frequency-modulated continuous wave (FMCW) radar systems.

BACKGROUND

[0003] One form of distance measuring radar is based on frequency modulation, in which a frequency of a returned signal is compared to that of the transmitted signal. This technique can be used in continuous wave radar and is used, for example, in vehicle applications such as adaptive cruise control (ACC), emergency braking, pedestrian detection, and 360-degree sensing, as well as industrial applications such as building automation, people counting, and surveillance. In these systems a carrier signal is frequency modulated in a predetermined way to provide a chirp and transmitted to a region of interest. A received reflection from the region of interest can be continuously compared to the transmitted chirp using a beat frequency modulator that produces a frequency tone representing a time delay between transmission of the chirp and reception of the reflection. This can be determined from the known frequency modulation of the chirp. A distance that the reflected signal travelled, which is twice the distance of the object from which the signal had reflected from the radar system, can be determined from this time delay.

SUMMARY OF THE INVENTION

[0004] In accordance with an aspect of the present disclosure, a system includes a radar sensor and a processor. The radar sensor is configurable to receive radar signals. The processor is configurable to generate a Doppler Fourier Transform (Doppler-FT) representation of the radar signals, and perform processing to adjust a noise floor in the Doppler-FT representation from a first level to a second level that is lower than the first level. The processing to adjust the noise floor is based on a velocity associated with a first signal peak visible in the Doppler-FT representation when the noise floor is at the first level, and a second signal peak not visible in the Doppler-FT representation when the noise floor is at the first level is visible in the Doppler-FT representation when the noise floor is at the second level.

[0005] In accordance with another aspect of the present disclosure, a non-transitory computer readable medium stores a set of instructions executable by a processor. The set of instructions comprises instructions for generating a Doppler Fourier Transform (Doppler-FT) representation based on received radar signals; and instructions for performing processing to adjust a noise floor in the Doppler-FT representation from a first level to a second level that is lower than the first level

based on a velocity associated with a first signal peak visible in the Doppler-FT representation when the noise floor is at the first level, such that a second signal peak not visible in the Doppler-FT representation when the noise floor is at the first level is visible in the Doppler-FT representation when the noise floor is at the second level.

[0006] Other aspects and features are described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 illustrates a frequency-modulated continuous wave (FMCW) radar system;

[0008] FIG. 2 is a graphical representation of a Doppler-FFT performed on range-gate data before correction for phase noise induced by a variable intrachirp interval;

[0009] FIG. 3 is a graphical representation of a Doppler-FFT performed on the same range gate data as FIG. 2 after the range-gate data has been corrected for the phase noise caused by dither that is associated with the strongest signal peak;

[0010] FIG. 4 illustrates one example of a signal processing component for processing frequency-modulated continuous wave (FMCW) radar signals;

[0011] FIG. 5 illustrates one example of a method for detecting moving objects a region of interest;

[0012] FIG. 6 illustrates another example of a method for detecting moving objects within a region of interest; and

[0013] FIG. 7 is a schematic block diagram illustrating an exemplary system of hardware components capable of implementing examples of the systems and methods disclosed in FIGS. 1-6, such as the digital signal processor illustrated in FIG. 4.

DETAILED DESCRIPTION OF THE INVENTION

[0014] One method of addressing interference between radar systems in close proximity is adding a known variability to the spacing between radar chirps, referred to as “dither.” Unfortunately, the added dither increases the phase noise when comparing the phase of a signal return for a given target between two chirps during a Doppler analysis, causing each moving object in a scanned region to produce a “phase skirt” of noise. The severity of the issue increases with the strength of the signal, and a sufficiently strong signal in the Doppler analysis can raise the noise floor in a Doppler analysis above the peaks representing weaker signals. As a result, objects can be lost in the phase noise, reducing the effectiveness of the radar system.

[0015] To mitigate the effects of dither on the Doppler analysis, the systems and methods herein use a correction value calculated from a velocity associated with a strong signal peak. This correction value reduces the phase noise caused by the strong signal peak, allowing weaker signals to be detected. Accordingly, a first Doppler analysis is performed on a set of range-gate data to provide a velocity measurement for an object represented by the strong signal. A correction value is calculated from the identified velocity and applied to the range-gate data, and a second Doppler analysis is performed on the corrected range-gate data to detect any weaker signals hidden by the noise.

[0016] FIG. 1 illustrates a frequency-modulated continuous wave (FMCW) radar system **100**. The system **100** includes a transmitter **102** that transmits a FMCW electromagnetic radiation (EM) signal at a region of interest. It will be appreciated that the transmitter can include one or multiple antennas. In one implementation, the frequency-modulated continuous wave EM signal is a “chirp” signal having a frequency that increases with linearly with time, although it will be appreciated that the system **100** can utilize any appropriate signal in which the frequency varies with time in a known fashion. The transmitter **102** can provide the chirp signals in sets, referred to as frames. In one example, each frame contains sixty-four chirps, and the transmitter **102** transmits twenty frames per second.

[0017] A radar sensor module **104** receives electromagnetic radiation reflected from the region of interest for each chirp. It will be appreciated that the transmitter can include one or multiple antennas. The radar sensor module **104** generates a signal representing the received electromagnetic radiation by mixing the received signal with a signal representing the transmitted electromagnetic radiation at an analog mixer and sampling a resulting intermediate frequency (IF) signal, or beat signal, at an analog-to-digital converter (ADC) to provide a time series of IF samples for each of one or more of antennas. Via the mixing process, the transmitted FMCW signal is compared to the return signal to determine a difference in frequency between the transmitted and returned signals. The samples are provided to a signal processor **106**. In one implementation, the one or more antennas are a plurality of virtual antennas, with outputs extracted via digital signal processing from the outputs of an array of physical antennas. In such a case, a time series of IF samples is generated for each of the virtual antennas. An example of a radar system using a time division multiplexed multiple input and multiple output (TDM-MIMO) implementation of a virtual antenna array can be found in TI Application Report SWRA554A, available at <http://www.ti.com/lit/an/swra554a/swra554a.pdf>, which is hereby incorporated by reference.

[0018] The signal processor **106** evaluates the time series of IF samples to detect objects within the region of interest. It will be appreciated that the signal processor can be implemented, for example, as a dedicated microprocessor. The signal processor provides a frequency domain representation of each time series of IF samples as values for a set of range bins representing distance from the radar sensor module **104**. Since the frequency modulation pattern is known, the change in frequency between the signals at a given time, as represented by the frequency of the IF signal, can provide the time of flight for the signal, and thus the range information from which the returned signal was reflected. Accordingly, a discretized representation of the return strength over range, referred to herein as “range bin values”, can be determined. Accordingly, for each chirp signal, a set of range bin values can be determined for each of the plurality of antennas.

[0019] In one implementation, the set of range bin values is determined by averaging a set of IF samples from the IF signal for the chirp to provide a DC component and subtracting the DC component from each sample to provide a set of DC-compensated samples. The DC-compensated samples are then subjected to a Fast Fourier Transform (FFT) to provide the set of range bin values. In some applications, the DC-compensation can be preceded by a windowing operation, instead of using all samples received after a given chirp. Accordingly, after a frame of chirps has been evaluated, the resulting data can be conceptualized as a two-dimensional data structure, in which each sample value represents a given chirp and range bin.

[0020] The signal processing element **106** then evaluates the frame across chirps to detect motion of any detected objects within the region of interest. To this end, the signal processing element **106** performs a Doppler FFT on the range data to detect differences in phase for a given range bin across the frame. The change in phase between two chirps represents a change in the position of the object during the time between the two chirps, and is thus indicative of a velocity of the object. For a change in phase, $\Delta\phi$, in a system using a radar of wavelength λ , a difference in the position of the object, Δd , can be determined as:

[00001]
$$d = \frac{\lambda \Delta\phi}{4} \quad \text{Eq. 1}$$

[0021] In some environments, it is possible that multiple radar systems can operate in reasonably close proximity. For example, in applications in which the radar sensors are employed in vehicle safety systems, it is quite likely that multiple vehicles employing similar systems can be on a same road or in a same parking lot. As a result, signals sent by a transmitter associated with one radar system could be received at a radar sensor module associated with another radar system, causing potentially erroneous results. To avoid interference between systems, dither can be added to the interchirp timing in each frame when sending chirps from the transmitter **102**. Accordingly, the interval between chirps varies according to a known pattern that is different across various radar

systems.

[0022] Since the dither pattern is known, the signal processor **106** can compensate for the differences in interchirp timing in generating the range bin data. Unfortunately, the dither increases phase noise in the Doppler FFT, which can mask any weaker signals, representing other moving objects, in the Doppler FFT. To this end, the signal processor **106** can identify a strong signal return, representing a detected object, in the Doppler FFT data, that is, a signal return that exceeds the noise floor. A velocity of the detected object can be determined from the Doppler FFT, and a correction factor can be calculated from the determined velocity. This correction value can be applied to the range bin data and a second Doppler FFT can be performed on the corrected data to detect any signals that were below the noise floor in the first Doppler FFT.

[0023] FIG. **2** is a graphical representation **200** of a Doppler-FFT performed on range-gate data before correction for phase noise induced by a variable intrachirp interval. A horizontal axis **202** represents velocity as a set of velocity bins, and a vertical axis **204** represents signal amplitude in decibels. A given velocity bin, k , represents a velocity, v , in a system using a radar of wavelength λ as $v = k\lambda/2NT_{\text{sub.c}}$, where N is the total number of bins in the Doppler FFT and $T_{\text{sub.c}}$ is the interchirp time interval. It will be appreciated that only one signal peak **206** is visible above a noise floor **208** of the Doppler-FFT, at approximately 120 m/s.

[0024] FIG. **3** is a graphical representation **300** of a Doppler-FFT performed on the same range gate data as FIG. **2** after the range-gate data has been corrected for the phase noise caused by dither that is associated with the strongest signal peak **302**. Specifically, a correction value calculated from the velocity associated with the strongest signal peak **302** was calculated and applied to the range-gate data. Once again, a horizontal axis **304** represents a velocity bin index, and a vertical axis **306** represents signal amplitude in decibels. After the correction, the noise floor **308** has been reduced by approximately twenty-five decibels relative to the noise floor **208** in FIG. **2**, and a second peak **310**, at around velocity bin index **60**, is now visible. It will be appreciated that the second peak **310** represents an object that was masked by the noise floor prior to the correction, which could result in suboptimal output from a vehicle safety system or other application.

[0025] FIG. **4** illustrates one example of a signal processing component **400** for processing frequency-modulated radar signals. For example, the signal processing component **400** can be used to provide the function of the signal processor **106** of FIG. **1**. In the illustrated example, the signal processing component **400** receives a time series of intermediate frequency (IF) signal samples. An IF, or beat, signal between the transmitted chirp and the received signal from each chirp is generated at a mixer and sampled at an analog-to-digital converter (ADC) during the chirping time. The samples for each chirp are processed in a range processing block **402**.

[0026] The range processing block **402** performs a Fast Fourier Transform (FFT) to provide, for each antenna, a set of values representing a strength of the reflected signal from each of a plurality of discrete ranges from the antenna. In one implementation, a windowing function can be applied to the input samples for the FFT prior to generating the FFT. A direct current (DC) component in the input signal samples can be estimated by averaging the samples in each chirp, and subtracted from each sample in the chirp. If the windowing function is used, any estimation and compensation for DC is applied after the windowing operation. In the illustrated implementation, the FFT for each chirp for each antenna is stored in a buffer as a set of range bin values. Given that there are multiple, $N_{\text{sub.c}}$, chirps in each frame and a defined number of range bins, $N_{\text{sub.f}}$, this provides a two-dimensional data structure with $N_{\text{sub.f}} \times N_{\text{sub.c}}$ entries.

[0027] The set of range-bin values is provided to an object detection component **404** that locates any objects within the region of interest. In one example, a one-dimensional constant false alarm rate (CFAR) algorithm can be used to detect objects at each range bin. In one implementation, an input for the CFAR algorithm is prepared by averaging the sample values, $x_{\text{sub.n,k,p}}$, over all the chirps within a frame and all the virtual receive antennas such that a n th element, s_o , of a range profile vector, $s = [s_{\text{sub.1}}, s_{\text{sub.2}}, \dots, s_{\text{sub.N.sub.r}}]$, is determined as:

$$[00002] \quad s_n = \frac{1}{N_c N_a} \cdot \text{Math.}_{p=1}^{N_a} \cdot \text{Math.}_{k=1}^{N_c} \cdot \text{Math.}_{n,k,p} \cdot \text{Math.}^2 \quad \text{Eq. 2}$$

[0028] Once the range profile vector is determined, a one-dimensional CFAR algorithm can be performed on the range profile vector, s , to determine which range bins are likely to contain objects, referred to herein as “detected range bins”. In a CFAR algorithm, values in the range profile vector that are peaks, that is, rise above the surrounding elements in the array (e.g., vector or matrix) as well as a background level, can be selected as detected range bins.

[0029] Once a set of detected range bins have been located, they are provided to a Doppler processing element **406** as a set of range-gate data. Specifically, range-gate data is provided to the Doppler processing element as sets of phasors representing phase differences between adjacent chirps in a radar frame for each detected range bin in the range FFT. The range-gate data for a given range-bin can be represented as:

$$[00003] \quad x_i(n) = \text{Math.}_{i=1}^m A_i e^{j(\phi_i(n) + \phi_i(n))} \quad \text{Eq. 3} \quad [0030] \text{ where,}$$

[00004] $\phi_i(n) = \frac{4 \cdot v_i T_c}{\lambda}$ is an ideal phase difference the $n-1$.sup.th and the n .sup.th chirp in a frame for an i .sup.th target in the range bin, that is a phase difference for the target in the absence of phase noise induced by dither, $T_{\text{sub.C}}$ is an interchirp interval before dither is applied,

$$[00005] \quad \phi_i(n) = \frac{4 \cdot v_i \Delta t_n}{\lambda}$$

is the phase noise induced by the dither in the timing between the $n-1$.sup.th and the n .sup.th chirp in a frame for an i .sup.th target, $A_{\text{sub.i}}$ is an amplitude of an i .sup.th target, m is a number of targets, Δt_n is the dither in the timing between the $n-1$.sup.th and the n .sup.th chirp in a frame and λ is a wavelength associated with a carrier frequency of the transmitted chirp.

[0031] The Doppler processing element **406** performs a Doppler FFT on the range-gate data to produce a velocity for each of the targets. The Doppler FFT represents the targets in the region of interest as amplitudes of a signal return across velocity, and thus a velocity can be determined for each target above the noise floor. One or more velocity values, along with the range-gate data, can be provided to a Doppler correction element **408** that computes corrections value from the velocity value or values and applies it to the range-gate data to provide a corrected set of range gate data. Since the phase noise associated with a given target, $\phi_{\text{sub.i}}(n)$, is a function of the velocity of the target, the correction value can simply be determined for each chirp as an multiplicative inverse of the expected phase noise for the identified velocity and multiplied with the original range gate value to provide the corrected phase gate data.

[0032] To provide a simplified example, assume that a given range bin includes two targets, such that the range-gate data for a given range bin can be represented as:

$$[00006] \quad x(n) = A_1 e^{j(\phi_1(n) + \phi_1(n))} + A_2 e^{j(\phi_2(n) + \phi_2(n))} \quad \text{Eq. 4}$$

[0033] Assuming that the amplitude associated with the first target is the larger of the two signal amplitudes, a set of correction values, CV, for the phase noise caused by the first target can be calculated from a velocity, $v_{\text{sub.1}}$, associated with the first target and determined from the Doppler FFT as

$$[00007] \quad CV = e^{j \phi_1(n)} = e^{\frac{4 \cdot j v_{\text{sub.1}} \Delta t_n}{\lambda}} \quad \text{Eq. 5}$$

[0034] Each correction value represents a phasor rotation on the range-gate data. The correction value can be multiplied with the range-gate data for each chirp to provide corrected range gate values as:

$$[00008] \quad x'(n) = A_1 e^{j \phi_1(n)} + A_2 e^{j(\phi_2(n) + \phi_2(n) - \phi_1(n))} \quad \text{Eq. 6}$$

[0035] As seen in equation 6, the phase noise has been eliminated from the stronger target (first target with amplitude $A_{\text{sub.1}}$), which is the dominant contributor to the noise floor. The Doppler processing element **406** can perform a Doppler FFT on the corrected range-gate data to provide a Doppler FFT with a reduced noise floor, such that a signal peak associated with the second target can be detected and used to determine a velocity for the second target.

[0036] In one implementation, the Doppler processing element **406** will only provide the range-gate data to the Doppler correction element **408** if one of the signal amplitudes is above a threshold value. The threshold value can be determined according to known weak target detection requirements of the radar system. Alternatively or additionally, the Doppler processing element **406** can provide the range-gate data to the Doppler correction element **408** iteratively to ensure that the phase noise for all targets having an amplitude above a threshold value is corrected and all weaker signals are detected than can practically be detected for a given radar sensor.

[0037] In view of the foregoing structural and functional features described above, example methods will be better appreciated with reference to FIGS. **5** and **6**. While, for purposes of simplicity of explanation, the example methods of FIGS. **5** and **6** are shown and described as executing serially, it is to be understood and appreciated that the present examples are not limited by the illustrated order, as some actions could in other examples occur in different orders, multiple times and/or concurrently from that shown and described herein. Moreover, it is not necessary that all described actions be performed to implement a method.

[0038] FIG. **5** illustrates one example of a method **500** for detecting objects within a region of interest. At **502**, range-gate data representing the position of an object within a region of interest is generated at a radar sensor. In one example, the radar sensor is a frequency modulated continuous wave radar. At **504**, a first Doppler-FFT is performed on the range-gate data to provide a first set of Doppler-FFT data. At **506**, a velocity is determined from the first set of Doppler-FFT data. Specifically, the velocity is determined from a signal amplitude in the first set of Doppler-FFT data that exceeds a noise floor in the Doppler-FFT data. In practice, the strongest signals produce the largest amount of dither-induced phase noise, and thus the velocity associated with a highest signal amplitude can be selected.

[0039] At **508**, a correction value for the range-gate data is determined from the identified velocity. The correction value is determined as to correct the range-gate data for phase noise caused by a variable interchirp interval, such as applied dither. In one example, a set of correction values, each associated with a chirp in a frame of radar returns, are determined as

$$[00009] e^{\frac{4jv_{\subscript{1}}n}{\Delta_{\subscript{1}}}},$$

where $v_{\subscript{1}}$ is the velocity associated with the signal amplitude that exceeds the noise floor, $\Delta_{\subscript{1}}$ is the dither in the timing between the $n-1_{\subscript{sup}}.th$ and the $n_{\subscript{sup}}.th$ chirp in the frame, and λ is a wavelength associated with a carrier frequency of the transmitted chirp.

[0040] At **510**, the range-gate data is modified using the determined correction value to provide modified range-gate data. In one example, a phase rotation is applied to the range-gate data based on the determined correction value. Once the data is corrected a second Doppler-FFT is performed on the modified range-gate data at **512** to provide a second set of Doppler-FFT data. In one implementation, the method **500** can be repeated to correct the modified range-gate data for additional strong peaks. For example, the method can be repeated until the amplitude associated with each peak in the Doppler-FFT data is below a threshold value. In another example, to save processing resources, the calculation and application of the correction value at **508**, **510**, and **512** are performed only if the highest signal amplitude in the first Doppler-FFT at **504** exceeds a threshold value.

[0041] FIG. **6** illustrates another example of a method **600** for detecting objects within a region of interest. At **602**, range-gate data representing the position of an object within a region of interest is generated at a radar sensor, such as a frequency modulated continuous wave radar. At **604**, a Doppler-FFT is performed on the range-gate data to provide a set of Doppler-FFT data. At **606**, it is determined if a highest signal amplitude in the set of Doppler-FFT data exceeds a threshold value. Essentially, the comparison at **606** determines if there is a signal of sufficient strength to mask other, weaker signals with its phase noise skirt, and the threshold can be selected according to weak target detection requirements of the radar sensor. If the highest signal amplitude in the Doppler-

FFT data does not exceed the threshold (N), the Doppler-FFT data is accepted at **608**.

[0042] If the highest signal amplitude in the Doppler-FFT data exceeds the threshold (Y), the method proceeds to **610**, where a velocity is determined from a signal amplitude in the set of Doppler-FFT data that exceeds a noise floor. At **612**, a correction value for the range-gate data is determined from the identified velocity. The correction value is determined as to correct the range-gate data for phase noise caused by a variable interchirp interval, such as applied dither. In one example, a set of correction values, each associated with a chirp in a frame of radar returns, are determined as

$$[00010]e^{\frac{4jv_{\text{sub}.1}n}{\Delta_{\text{sub}.n}\lambda}}$$

where $v_{\text{sub}.1}$ is the velocity associated with the signal amplitude that exceeds the noise floor, $\Delta_{\text{sub}.n}$ is the dither in the timing between the $n-1$.sup.th and the n .sup.th chirp in the frame and λ is a wavelength associated with a carrier frequency of the transmitted chirp. At **614**, the range-gate data is modified using the determined correction value. In one example, a phase rotation is applied to the range-gate data based on the determined correction value. The method then returns to **604** to perform a Doppler-FFT on the modified range-gate data.

[0043] FIG. 7 is a schematic block diagram illustrating an exemplary system **700** of hardware components capable of implementing examples of the systems and methods disclosed in FIGS. 1-6, such as the digital signal processor **400** illustrated in FIG. 4. The system **700** can include various systems and subsystems. The system **700** can be a personal computer, a laptop computer, a workstation, a computer system, an appliance, an application-specific integrated circuit (ASIC), a server, a server blade center, a server farm, etc. In one implementation, the system **700** is a microprocessor, and it will be appreciated that in this implementation, some of the components **702**, **704**, **706**, **708**, **710**, **712**, **714**, **716**, and **718** may not be necessary. In one example, the digital signal processor **400** can be implemented as an integrated circuit. Examples of integrated circuit hardware that could be adapted for this purpose can be found in TI datasheet SWRS203A, available at [http://www.ti.com/general/docs/datasheetdiagram.tsp?](http://www.ti.com/general/docs/datasheetdiagram.tsp?genericPartNumber=AWR1642&diagramId=SWRS203A)

[genericPartNumber=AWR1642&diagramId=SWRS203A](http://www.ti.com/general/docs/datasheetdiagram.tsp?genericPartNumber=AWR1642&diagramId=SWRS203A), and TI datasheet SWRS202A, available at [http://www.ti.com/general/docs/datasheetdiagram.tsp?](http://www.ti.com/general/docs/datasheetdiagram.tsp?genericPartNumber=AWR1443&diagramId=SWRS202A)

[genericPartNumber=AWR1443&diagramId=SWRS202A](http://www.ti.com/general/docs/datasheetdiagram.tsp?genericPartNumber=AWR1443&diagramId=SWRS202A), each of which are hereby incorporated by reference.

[0044] The system **700** can includes a system bus **702**, a processing unit **704**, a system memory **706**, memory devices **708** and **710**, a communication interface **712** (e.g., a network interface), a communication link **714**, a display **716** (e.g., a video screen), and an input device **718** (e.g., a keyboard and/or a mouse). The system bus **702** can be in communication with the processing unit **704** and the system memory **706**. The additional memory devices **708** and **710**, such as a hard disk drive, server, stand-alone database, or other non-volatile memory, can also be in communication with the system bus **702**. The system bus **702** interconnects the processing unit **704**, the memory devices **706-710**, the communication interface **712**, the display **716**, and the input device **718**. In some examples, the system bus **702** also interconnects an additional port (not shown), such as a universal serial bus (USB) port.

[0045] The processing unit **704** can be a computing device and can include an application-specific integrated circuit (ASIC). The processing unit **704** executes a set of instructions to implement the operations of examples disclosed herein. The processing unit can include a processing core.

[0046] The additional memory devices **706**, **708**, and **710** can store data, programs, instructions, database queries in text or compiled form, and any other information that can be needed to operate a computer. The memories **706**, **708** and **710** can be implemented as computer-readable media (integrated or removable) such as a memory card, disk drive, compact disk (CD), or server accessible over a network. In certain examples, the memories **706**, **708** and **710** can comprise text, images, video, and/or audio, portions of which can be available in formats comprehensible to

human beings. Additionally or alternatively, the system **700** can access an external data source or query source through the communication interface **712**, which can communicate with the system bus **702** and the communication link **714**.

[0047] In operation, the system **700** can be used to implement one or more parts of a system in accordance with the present invention. Computer executable logic for implementing the system resides on one or more of the system memory **706**, and the memory devices **708**, **710** in accordance with certain examples. The processing unit **704** executes one or more computer executable instructions originating from the system memory **706** and the memory devices **708** and **710**. The term “computer readable medium” as used herein refers to any medium that participates in providing instructions to the processing unit **704** for execution, and it will be appreciated that a computer readable medium can include multiple computer readable media each operatively connected to the processing unit.

[0048] What have been described above are examples. It is, of course, not possible to describe every conceivable combination of components or methodologies, but one of ordinary skill in the art will recognize that many further combinations and permutations are possible. Accordingly, the disclosure is intended to embrace all such alterations, modifications, and variations that fall within the scope of this application, including the appended claims. As used herein, the term “includes” means includes but not limited to, the term “including” means including but not limited to. The term “based on” means based at least in part on. Additionally, where the disclosure or claims recite “a,” “an,” “a first,” or “another” element, or the equivalent thereof, it should be interpreted to include one or more than one such element, neither requiring nor excluding two or more such elements.

[0049] What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications, and variations that fall within the scope of the appended claims.

Claims

1. A system comprising: a radar sensor configurable to receive radar signals; and a processor configurable to: generate a Doppler Fourier Transform (Doppler-FT) representation of the radar signals, and perform processing to adjust a noise floor in the Doppler-FT representation from a first level to a second level that is lower than the first level, wherein: the processing to adjust the noise floor is based on a velocity associated with a first signal peak visible in the Doppler-FT representation when the noise floor is at the first level, and a second signal peak not visible in the Doppler-FT representation when the noise floor is at the first level is visible in the Doppler-FT representation when the noise floor is at the second level.
2. The system of claim 1, wherein the first signal peak is a dominant peak in the Doppler-FT representation.
3. The system of claim 1, wherein the processor includes: Doppler processing circuitry configurable to receive range data that is based on the radar signals and generate the Doppler-FT representation having the noise floor at the first level based on the range data; and correction circuitry configurable to estimate a phase noise associated with the first signal peak, and modify the range data based on the estimated phase noise to provide modified range data.
4. The system of claim 3, wherein the Doppler-FT representation is a first Doppler-FT representation having the noise floor at the first level, the Doppler processing circuitry configurable to generate a second Doppler-FT representation based on the modified range data, the second Doppler-FT representation having the noise floor at the second level.

5. The system of claim 3, wherein the radar sensor is further configurable to transmit radar chirp signals with dithering, and the phase noise associated with the first signal peak, which the correction circuitry is configurable to estimate, is induced by the dithering.
6. The system of claim 5, wherein the correction circuitry is configurable to estimate the phase noise associated with the first signal peak based on the velocity associated with the first signal peak and a timing of the dithering.
7. The system of claim 6, wherein the correction circuitry is further configurable to multiply the range data by a conjugate phasor corresponding to the estimated phase noise to provide the modified range data.
8. The system of claim 7, wherein, in the modified range data, the estimated phase noise is removed from a representation of the first signal peak.
9. The system of claim 8, wherein, in the modified range data, phase noise associated with the second signal peak is increased based on the estimated phase noise of the first signal peak.
10. The system of claim 1, further comprising a memory coupled to the radar sensor and the processor, the memory configurable to store at least one of data input to the processor and data output by the processor.
11. A non-transitory computer readable medium storing a set of instructions executable by processing circuitry, the set of instructions comprising: instructions for generating a Doppler Fourier Transform (Doppler-FT) representation based on received radar signals; and instructions for performing processing to adjust a noise floor in the Doppler-FT representation from a first level to a second level that is lower than the first level based on a velocity associated with a first signal peak visible in the Doppler-FT representation when the noise floor is at the first level, such that a second signal peak not visible in the Doppler-FT representation when the noise floor is at the first level is visible in the Doppler-FT representation when the noise floor is at the second level.
12. The non-transitory computer readable medium of claim 11, wherein the first signal peak is a dominant peak in the Doppler-FT representation.
13. The non-transitory computer readable medium of claim 11, wherein the processing circuitry includes Doppler processing circuitry and correction circuitry, in which: the instructions for generating the Doppler-FT representation are executable by the Doppler processing circuitry and include instructions for generating the Doppler-FT representation having the noise floor at the first level based on range data that is based on the received radar signals; and the instructions for performing processing to adjust the noise floor in the Doppler-FT representation are executable by the correction circuitry.
14. The non-transitory computer readable medium of claim 13, wherein the instructions for performing processing to adjust the noise floor in the Doppler-FT representation include instructions for estimating a phase noise associated with the first signal peak and instructions to modify the range data based on the estimated phase noise to provide modified range data.
15. The non-transitory computer readable medium of claim 14, wherein the Doppler-FT representation is a first Doppler-FT representation having the noise floor at the first level, and the set of instructions further comprise: instructions, executable by the Doppler processing circuitry, for generating a second Doppler-FT representation based on the modified range data, the second Doppler-FT representation having the noise floor at the second level.
16. The non-transitory computer readable medium of claim 14, wherein the set of instructions further comprise: instructions for transmitting radar chirp signals with dithering, the phase noise associated with the first signal peak being induced by the dithering.
17. The non-transitory computer readable medium of claim 16, wherein the set of instructions for estimating the phase noise associated with the first signal peak includes instructions for estimating the phase noise associated with the first signal peak based on the velocity associated with the first signal peak and a timing of the dithering.
18. The non-transitory computer readable medium of claim 17, wherein the set of instructions to

modify the range data based on the estimated phase noise include instructions for multiplying the range data by a conjugate phasor corresponding to the estimated phase noise to provide the modified range data.

19. The non-transitory computer readable medium of claim 18, wherein the set of instructions further comprise instructions for removing the estimated phase noise associated with the first signal peak in the modified range data.

20. The non-transitory computer readable medium of claim 19, wherein the set of instructions further comprise instructions for increasing phase noise associated with the second signal peak in the modified range data based on the estimated phase noise.
